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DEPARTMENT OF ANAESTHESIA
BCUHB (West)

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

ESTABLISHING EPIDURAL ANAESTHESIA FOR LABOUR ANALGESIA

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Author(s):	Dr Gareth Roberts	**Author(s) title:**	Consultant Anaesthetist
Responsible dept / director:	Department of Anaesthesia, BCUHB (West) Dr Linda Warnock Clinical Lead of Anaesthesia, BCUHB (West)		
Approved by:	Department of Anaesthesia, BCUHB (West)		
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1. Introduction and Purpose

Epidural anaesthesia has been available in the UK for labour analgesia since the late 1960's. It is a more effective form of analgesia compared with opiates and Entonox. However it is associated with risks such as post dural puncture headache (1:100) and permanent nerve damage (1:24,000 at 6 months).

2. Scope

This guideline is intended as a reference for anaesthetic trainees seeking a recommended technique for epidural anaesthesia for labour analgesia.

3. Points to consider

3.1 Patient Preparation

- Ensure the midwife has provided the patient with the Epidural Information Card [2]
- Full preoperative assessment of the patient
- Indications for epidural analgesia include :
    i. Maternal request
    ii. Multiple pregnancy or malpresentation (ie. increased risk of operative delivery)
    iii. Pre-eclampsia
    iv. Maternal cardiovascular disease (eg. IHD or regurgitant valvular lesions)
    v. Morbid obesity (to try and avoid a GA if operative delivery necessary)
- Ensure no contraindication to epidural :
    i. Patient refusal
    ii. Lack of midwifery supervision
    iii. Thrombocytopaenia (platelet count < 80)
    iv. Coagulopathy (INR >1.4) or patient on anticoagulation therapy
    v. Uncorrected hypovolaemia
    vi. Local sepsis
    vii. Systemic sepsis

Criteria for the diagnosis of systemic sepsis :

- *Clinical suspicion of sepsis* :

History - e.g diarrhoea, vomiting, UTI, chest infection chorioamnionitis etc.

& Clinical signs - Heart rate >100 / minute
RR >20 / minute
+ Temp. <36 ºC or > 38 ºC *

or - *Temp. >38 ºC* in the absence of an obvious non-infective cause

or - *WCC > 30,000 / mm³* in the absence of an obvious non-infective cause

or - *CRP > 40* in the absence of an obvious non-infective cause[1]

*** All temperature measurements should be undertaken for accuracy using a temporal artery scanning temperature probe**

(NB. Chronic backpain does not contraindicate epidural anaesthesia, but scarring of the epidural space e.g following spinal surgery may limit the effectiveness of epidural analgesia and increase the risk of inadvertent dural puncture)

- Discussion with the patient of:

    a) Benefits of an epidural:

        i. Better quality labour analgesia than opiates or Entonox

        ii. Can be topped up to give anaesthesia for Caesarean Section, instrumental delivery, manual removal of placenta or repair of perineal tears

        iii. In pre-eclampsia it improves uteroplacental bloodflow and reduces plasma catecholamines and stress response with consequent better control of blood pressure

    b) Risks of an epidural:

        i. Post dural puncture headache (1:100)

        ii. Nausea and vomiting

iii. Failure to provide effective analgesia for labour (1:8)

Failure to produce sufficient analgesia on topping up for Caesarean Section thus requiring a conversion to general anaesthesia (1:20)

iv. Nerve damage (Temporary = 1:1,000,
Permanent = 1:24,000 incidence at 6 months)

c) Epidural procedure

d) Explanation of what sensations she will feel once epidural is functioning

- Obtain verbal consent for the epidural and document this on the epidural chart
- Discuss local anaesthetic delivery ie. Programmed Intermittent Bolus (PIB) plus Patient Controlled Epidural Analgesia (PCEA)

3.2 Equipment and Staff

- Ensure presence of trained midwife to monitor epidural block once established
- Check all equipment is present to perform epidural block
- Ensure metaraminol and ephedrine are at hand

3.3 Technique

- Site a **16G** (or 14G) iv cannula
- A routine fluid preload is not required, however if the parturient is dehydrated a preload volume of **500ml Hartmanns solution** should be administered. Do not preload in severe pre-eclampsia
- Place patient in **Sitting** (can be cross-legged) or lateral position
- Use full aseptic technique :

  a) Gown, mask and gloves

  b) Spray the patient's back twice from the mid thoracic area to the upper natal cleft, and over to the flanks using **Derma Spray** (0.5% chlorhexidine in 70% alcohol). Allow the alcohol to dry each time.

- Skin and ligament infiltration with 1% Lignocaine at **L3/4** or L4/5 interspace

- 16G Tuohy needle (8cm) used (11cm Tuohy needle available for super morbidly obese) and inserted just into ligament followed by removal of stillet

- Epidural space located using loss of resistance to saline technique. This is nowadays preferred to LOR to air. Remember to always use the LOR syringe and not an ordinary 10 ml syringe (high resistance)

- Epidural catheter is threaded and left 5 cm in the epidural space (7cm in morbid obesity)

- Check that no blood or CSF can be aspirated from the epidural catheter

- Epidural catheter secured with adhesive clip, Tegaderm and Mefix dressings

- Give a combined test dose and loading dose of 15mls 0.1% levobupivacaine + 2 micrograms fentanyl slowly over 2 minutes (dose to be taken (using a clean technique) from the standard epidural infusion mixture bag)

- If after 5 minutes no evidence of a subarachnoid block (high, dense block) or intravascular injection (perioral tingling, lightheadedness, tinnitus etc.) has appeared, then you are ready to commence the epidural infusion

- Check the CADD – Solis epidural infusion pump programme*, then connect the epidural infusion set to the epidural filter and commence the epidural infusion

- Perform routine haemodynamic measurements (pulse, NIBP) every 5 minutes for 20 minutes following the initial block establishing bolus, and subsequently following any bolus of local anaesthetic greater than 10 mls 0.1% Levobupivacaine, or its equivalent. Subsequently take readings every 30 minutes if the patient is cardiovascularly stable, otherwise more often.

- At 20 – 30 minutes post initial dose, assess level of sensory block (temperature) using Ethyl Chloride spray and motor block using the Bromage Scale. Aim for T10 block bilaterally with minimal motor blockade

- Document the epidural procedure on the front page of the epidural chart

- Prescribe the epidural local anaesthetic infusion as follows :

Drug : 0.1% Levobupivacaine + 2mcg/ml fentanyl

Route : Epidural

Dose : PIB = 10ml
PCEA = 5 – 7ml

Lockout : PIB = 60 minutes
PCEA = 20 minutes

- Return to the patient at regular intervals during labour to assess the quality of epidural analgesia and liaise with midwife regarding any problems with the epidural.

3.4 Monitoring

- Irrespective of the epidural pump program chosen (PIB / PCEA or continuous infusion) the monitoring regime outlined below should be adhered to[6].

- Perform routine haemodynamic measurements (pulse & NIBP) every 5 minutes for 20 minutes following the epidural administration of :

    a) The initial establishing local anaesthetic / opiate bolus

    b) Any bolus dose **greater than** 10mg Levobupivacaine (ie greater than 10 mls 0.1% Levobupivacaine or equivalent)

- At **30 minute intervals** during the maintenance of epidural analgesia the midwife should monitor pulse & blood pressure. This should be performed more frequently if the patient is not cardiovascularly stable

- At **hourly intervals** during the maintenance of epidural analgesia the midwife should monitor :

    a) Respiratory Rate

    b) Height of sensory block (cold sensation) – including noting any missed segments

    c) Motor block

    d) Sedation score

    e) Pain score

    f) Epidural site

    g) Degree of bladder distension (if urinary catheter not in situ)

    h) Pressure areas

3.5 Discontinuation of the epidural

- Following satisfactory conclusion of third stage of delivery and repair of any perineal tears, the midwife should :

    a) Stop the epidural infusion

    b) Disconnect the epidural infusion and cap the epidural filter

c) Aseptically (using sterile gloves) remove the epidural catheter and dressings and make sure the tip is intact. If the epidural catheter is stuck try flexing the patient more. If it still cannot be removed then the anaesthetist should be called. NEVER USE EXCESSIVE TRACTION TO REMOVE AN EPIDURAL CATHETER AS IT MAY BREAK

d) Midwife should document that the tip of the epidural catheter was intact

e) Place a plaster over the epidural site

f) Monitor offset of epidural block for 6 hours after removing the epidural catheter

g) Withhold enoxaparin dose for 4 hours following removal of the epidural catheter

DO NOT REMOVE THE EPIDURAL CATHETER IF THE PLATELET LEVEL < 80 OR THE PATIENT HAS A COAGULOPATHY (eg severe pre-eclampsia ,HELLP, APH or PPH)

3.6 Methods of epidural infusion delivery

- Programmed Intermittent Bolus (PIB)

  The epidural pump delivers a bolus of a set volume at a set interval e.g instead of a 10ml/hr infusion, the pump delivers a regular 10 ml bolus at hourly intervals. PIB results in a greater and more uniform spread of local anaesthetic compared to CEI due to higher injectate pressures. The quality of analgesia provided by PIB is similar to continuous epidural infusion (CEI), however this method of delivery of epidural analgesia has been shown to reduces local anaesthetic consumption & motor block, reduces the number of clinician top up boluses required and may improve patient satisfaction[5]. PIB can be used with or without a PCEA function.

- Patient Controlled Epidural Analgesia (PCEA)

  The patient initiates delivery of a bolus of local anaesthetic epidurally. The volume of the bolus and lockout interval is pre-programmed. As with PIB the quality of analgesia provided by PCEA is similar to CEI. This method of delivery of epidural analgesia has also been shown to reduces local anaesthetic consumption & motor block, reduces the number of clinician top up boluses required and may improve patient satisfaction[3,4]. Fewer but larger volume PCEA doses (> 5ml) have been found to provide better analgesia than more frequent smaller doses (due to the local anaesthetic spread effect). PCEA can be used with or without PIB or a continuous background infusion.

- Continuous Epidural Infusion (CEI)

  A continuous infusion of local anaesthetic / opiate mix is administered epidurally. CEI is associated with an increase in local anaesthetic consumption and motor block compared with an intermittent bolus techniques, due to a poorer spread of local anaesthetic[3,5].

  Therefore in summary, epidural infusion delivery methods in order of lowest LA consumption and best maternal satisfaction scores are as follows :

  PCEA + PIB > PCEA + background CEI > PCEA alone > CEI

  For this reason PIB + PCEA is therefore the preferred method of delivery of epidural LA in our maternity unit.

3.7 <u>CADD – Solis Epidural Infusion Pump Programmes</u>

The CADD – Solis pump has two obstetric Epidural infusion programmes :

a) Combined PIB & PCEA programme

b) Continuous infusion programme

- **Setting up the CADD – Solis epidural pump for PIB / PCEA Programme**

  Ensure external PCEA bolus administration button plugged in (Port on left side)

  Switch pump on (Grey button on right)

  Screen displays “Do you want to start a new patient ?” Press “Yes”

  The “Select therapy” menu is displayed

  Scroll down to “Obstetric PCEA / PIB”. Press select

  The “Select Qualifier” menu is displayed. Confirm “0.1% Levobupivacaine”

  The “Select Drug” menu is displayed. Confirm “Fentanyl 2mcg/ml”

  Enter security code “551”

  Check programme. Press “Yes”

Review pump settings : Intermittent bolus (PIB) : 10 ml (not adjustable)

Bolus Interval : 1 hour (not adjustable)

Next (PIB) bolus : 20 minutes (default) (Range : 0 – 4 hours)

PCEA dose : 5 ml (default) (Range : 5 – 7 ml)

PCEA lockout : 20 minutes (not adjustable)

Reservoir volume : 100ml (default) (Range : 0 – 1000)

To adjust any values choose “select”, then edit amount followed by “save”

Remember to reduce the reservoir volume if you remove local anaesthetic for the initial combined test and loading dose

Press Accept Value

Screen displays “Start pump ?” Press “Yes”

Please note that “Clinician bolus” function (under “Tasks”) is not available when using PCEA / PIB programme

“Reports” function will provide data on PIB and PCEA usage, time to next PIB and total LA / opiate mix consumption

To change epidural infusion programme from PCEA / PIB to continuous infusion or vice versa go to “Tasks” and press “ Start new protocol, same patient”

- **Setting up the CADD – Solis epidural pump for a Continuous infusion Programme**

Switch pump on (Grey button on right)

Screen displays “Do you want to start a new patient ?” Press “Yes”

The “Select therapy” menu is displayed

Scroll down to “Obstetric Epidural”. Press select

The “Select Qualifier” menu is displayed. Confirm “0.1% Levobupivacaine”

The “Select Drug” menu is displayed. Confirm “Fentanyl 2mcg/ml”

Enter security code “551”

Check programme. Press “Yes”

Review pump settings : Continuous rate : 10 ml (default)
(Normal range : 10 – 15 ml / hr
Absolute maximum : 20 ml hr)

Reservoir volume : 100ml (default) (Range : 0 – 1000)

To adjust any values choose “select”, then edit amount followed by “save”

Remember to reduce the reservoir volume if you remove local anaesthetic for the initial combined test and loading dose

Press “Accept” Values

Screen displays “Start pump ?” Press “Yes”

“Clinician bolus” function (under “Tasks”) is available when using continuous infusion programme. The settings for clinician (midwife) bolus are as follows :

Clinician (midwife) bolus volume : 10 ml (not adjustable)

Clinician (midwife) bolus lockout : 1 hour (not adjustable)

4. References

1. De Meeus JB, Pourrat O, Gombert J, Magnin G.C-reactive protein levels at the onset of labour and at day 3 post-partum in normal pregnancy. *Clin Exp Obstet Gynecol* 1998;25(1-2):9-11.

2. Epidural Information Card. Obstetric Anaesthetists Association 2008

3. Van der Vyver M, Halpern S and Joseph G. Patient Controlled epidural analgesia vs continuous infusion for labour analgesia : a meta-analysis. British Journal of Anaesthesia **89** (3): 459-65 (2002)

4. Halpern S, Carvalho B. Patient Controlled Epidural Analgesia for Labour. Anaesthesia & Analgesia **108**: 921-8 (2009)

5. Wong CA, Ratliff JT, Sullivan JT, Scavone BM, Toledo P, Mc Carthy RJ. A Randomised Comparison of Programmed Intermittent Epidural Bolus with Continuous Epidural Infusion for Labour Analgesia. Anaesthesia & Analgesia **102**: 904-9 (2006)

6. Appendix 2 Epidural Analgesia – Patient Controlled Local Operating Procedure (2012). Royal Hospital for Women, NSW, Australia